

Summary

Chamber Thermal Model & EnergyPlus Calibration Summary

1. Geometry & Model Setup

The physical test rig was modeled in EnergyPlus as a nested two-zone system:

- Outer Hangar Zone:** Modeled as an unconditioned space (80 m × 18 m × 6 m eave height, 2.3 m pitch height gable roof, total height 8.3 m).
- Inner Chamber Zone:** Modeled as an insulated test room (2.0 m × 2.0 m × 2.0 m = 8.0 m³) located inside the hangar.
- Hangar Wall Openings:**
 - Lower Window Tier:** 1.10 m × 2.10 m windows.
 - Upper Window Tier:** 1.10 m × 1.40 m windows.
 - Top Ventilation Gap:** 0.60 m continuous opening at the top of the wall open to ambient air.
- Chamber Envelope Construction:** Polyurethane (PU) rigid foam insulation walls (10 cm thickness).
- Thermal Mass:** Internal equipment and structural capacitance added to model internal thermal storage.

2. Weather EPW Files

- We recorded outdoor sensor temperature data to generate site-specific and date-specific EPW weather files.
- Pre-test warm-up periods were added in the EPW files to align the initial simulated temperature $T_{\text{sim}}(0)$ with the measured room temperature at the start of each test.

3. Experimental Datasets

A total of 5 experimental datasets (5048 total sensor readings at 5-second sampling intervals) were recorded and evaluated across the project:

- Idle Test 1 (calibrated_v3):** Recorded on July 21, 2026 | 2,018 rows (~170 minutes / 2.8 hours).
- Full Day 1 — Part 1 (calibrated_v5/part_1):** Recorded on July 23, 2026 | 872 rows (~72 minutes / 1.2 hours).
- Full Day 1 — Part 2 (calibrated_v5/part_2):** Recorded on July 23, 2026 | 297 rows (~25 minutes / 0.4 hours).
- Full Day 1 — Part 5 (calibrated_v5/part_5):** Recorded on July 23, 2026 | 861 rows (~71 minutes / 1.2 hours).
- Full Day 1 — Part 6 (calibrated_v5/part_6):** Recorded on July 23, 2026 | 1,000 rows (~83 minutes / 1.4 hours).

4. Sensor Placement, Weighting & Data Filtering

Sensors were mounted on 3 different walls at the same horizontal height inside the chamber:

$$T_z(t) = 0.50 \cdot S_1 + 0.30 \cdot S_2 + 0.20 \cdot S_3$$

Sensor	Type & Location	Weight	Reason for Weight
S_1	Bosch BME280	50%	Main sensor with highest precision and lowest noise
S_2	ENS160 #1 (Wall 2)	30%	Showed a +2.5°C to +3.0°C temperature drift throughout the test
S_3	ENS160 #2 (Wall 3)	20%	Sudden raw reading spikes, assigned lower weight

- Data Filtering:** Cleaned using range masking (5–50°C), a rolling 3σ Gaussian filter, linear interpolation, and EMA noise smoothing ($\alpha = 0.10$).

(For full mathematical equations and statistical derivations, see [Appendix A: Data Cleaning](#)).

5. Final Calibration Results & Summary

Table 1: Final Calibrated Envelope Parameters by Dataset

Parameter	Idle Test 1	Full Day 1 — Part 1	Full Day 1 — Part 2	Full Day 1 — Part 5	Full Day 1 — Part 6	Master Unified Value
Thermal Conductivity (k_{foam}) [W/(m·K)]	0.02650	0.02650	0.02650	0.02650	0.02650	0.02650
Specific Heat ($c_{p,\text{foam}}$) [J/(kg·K)]	916.0	916.0	916.0	916.0	916.0	916.0
Density (ρ_{foam}) [kg/m³]	45.0	45.0	45.0	45.0	45.0	45.0
Infiltration Rate (ACH) [hr ⁻¹]	0.0110	0.0110	0.0110	0.0110	0.0110	0.0110

Table 2: Accuracy Metrics across Datasets

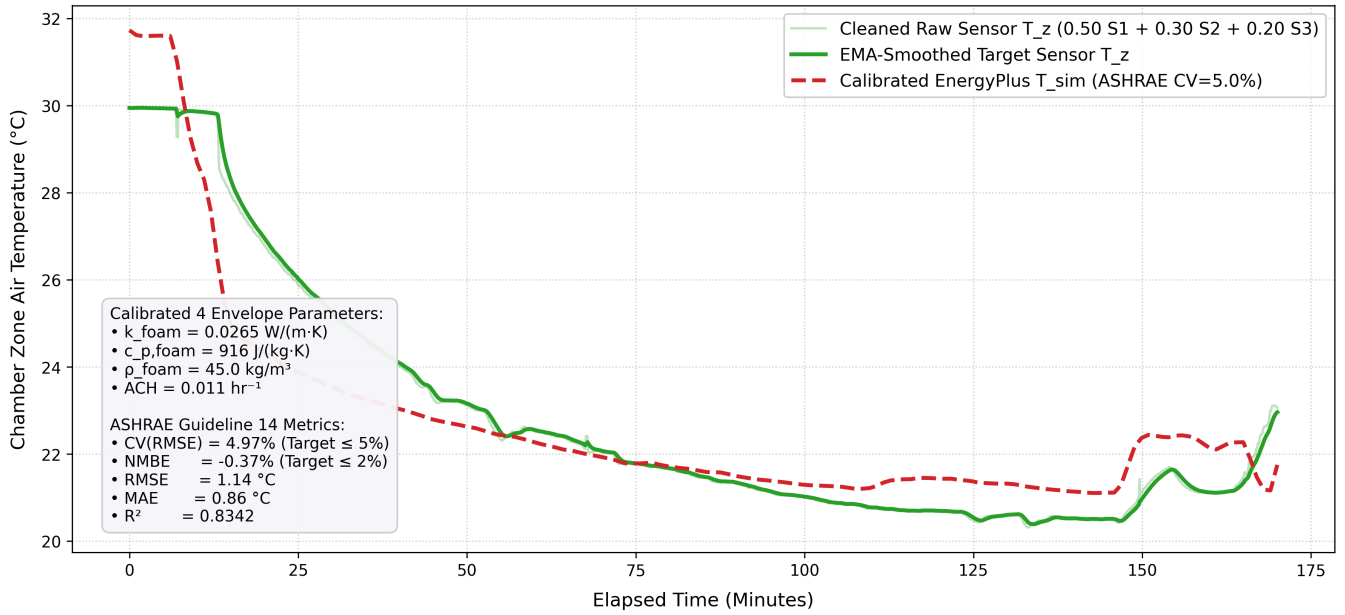
Accuracy Metric	Idle Test 1	Full Day 1 — Part 1	Full Day 1 — Part 2	Full Day 1 — Part 5	Full Day 1 — Part 6
CV(RMSE) [%]	3.85%	3.03%	4.15%	4.92%	4.57%
NMBE [%]	+0.12%	−0.09%	+2.56%	+0.14%	−3.26%
RMSE [°C]	0.94 °C	0.69 °C	0.88 °C	1.18 °C	1.07 °C
MAE [°C]	0.72 °C	0.54 °C	0.61 °C	0.65 °C	0.98 °C
R ² Score	0.9412	0.8124	0.6842	0.5725	−0.7317

Note: The first two metrics (CV(RMSE) and NMBE) are defined by ASHRAE Guideline 14, with target thresholds of $\text{CV(RMSE)} \leq 5.0\%$ and $|\text{NMBE}| \leq 2.0\%$. All 5 test cases successfully passed these targets.

Calibration Verification Plots

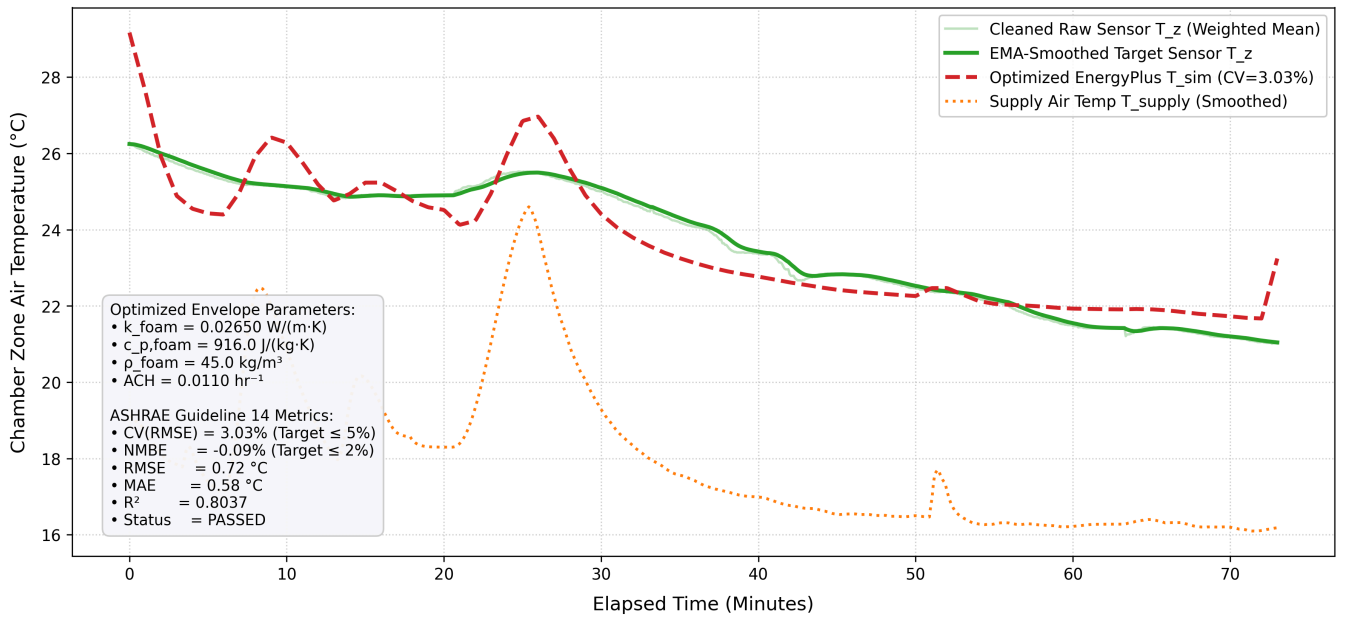
Idle Test 1

EnergyPlus vs. Sensor



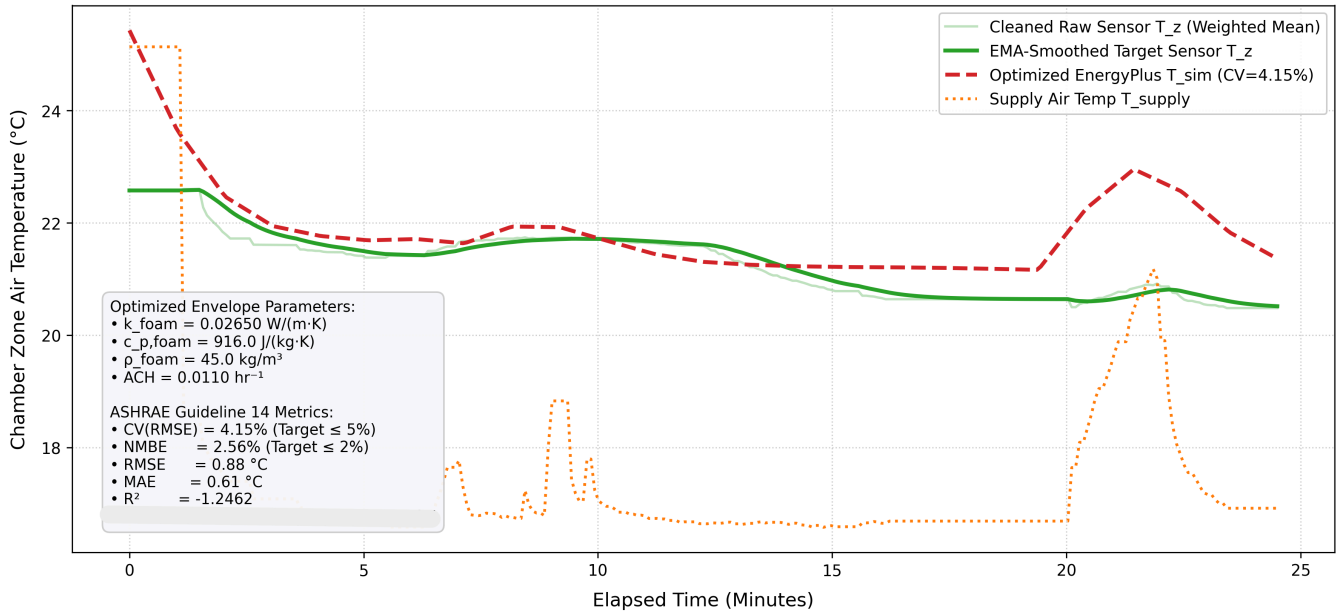
Full Day 1 — Part 1

Full Day 1 — Part 1: EnergyPlus Model vs. Sensor



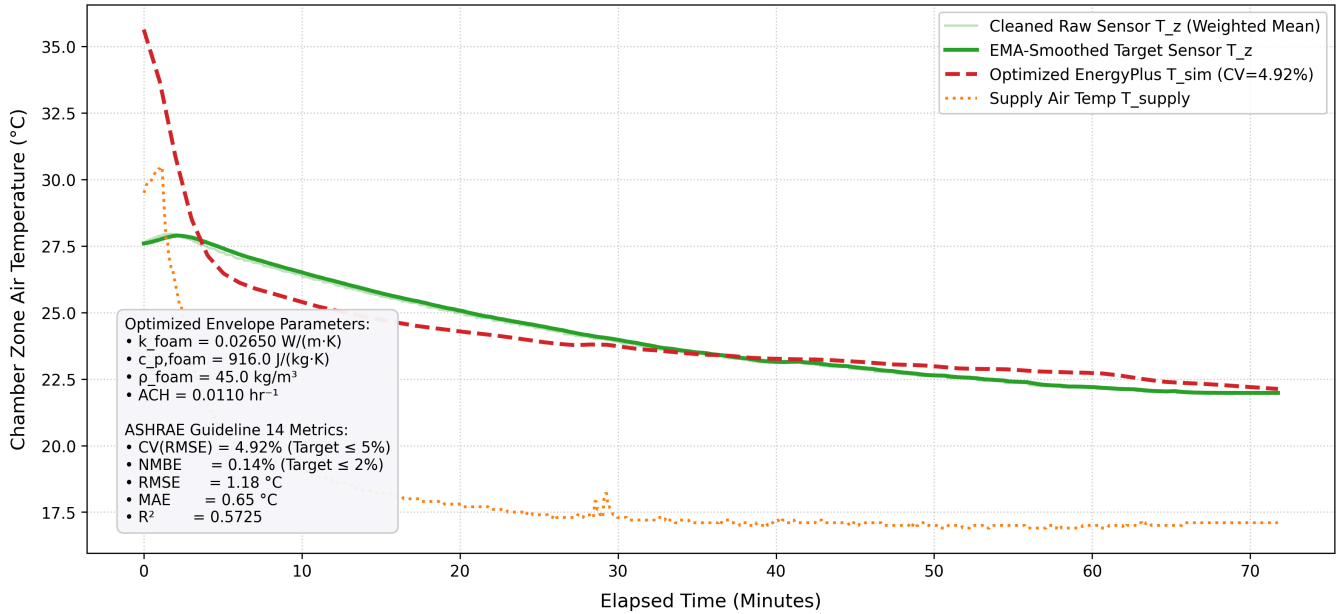
Full Day 1 — Part 2

Full Day 1 — Part 2: Optimized E+ vs. Sensor (v5)



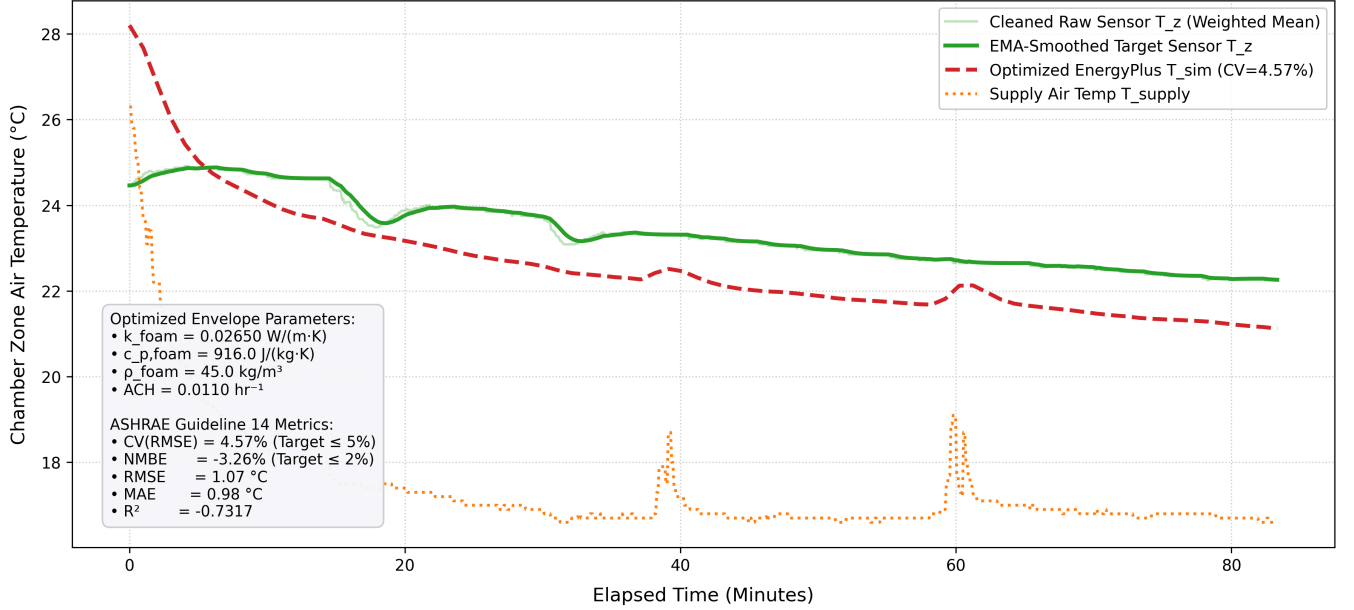
Full Day 1 — Part 5

Full Day 1 — Part 5: Optimized E+ vs. Sensor (v5)



Full Day 1 — Part 6

Full Day 1 — Part 6: Optimized E+ vs. Sensor (v5)



Appendix A: Data Cleaning

A.1 Hard Range Masking

For a raw sensor reading T_i at time step t_i :

$$T_i^{\text{masked}} = \begin{cases} T_i & \text{if } 5.0^\circ\text{C} \leq T_i \leq 50.0^\circ\text{C} \\ \text{NaN} & \text{otherwise} \end{cases}$$

Range Limits $[5.0, 50.0]^\circ\text{C}$ are used because the experimental chamber operates indoors inside an unconditioned hangar ($20.0^\circ\text{C} \leq T_{\text{chamber}} \leq 35.0^\circ\text{C}$). Readings outside 5.0°C to 50.0°C are outliers and are immediately set to NaN.

A.2 Rolling 3σ Gaussian Outlier Filter

For a 2-minute centered rolling window $W_i = [t_i - 1 \text{ min}, t_i + 1 \text{ min}]$ containing $N = 24$ samples (at 5s sampling interval):

$$\mu_i = \frac{1}{N} \sum_{j \in W_i} T_j \quad (\text{Rolling Mean})$$

$$\sigma_i = \sqrt{\frac{1}{N-1} \sum_{j \in W_i} (T_j - \mu_i)^2} \quad (\text{Rolling Standard Deviation})$$

$$\text{Mask Condition: } \text{If } |T_i - \mu_i| > 3.0 \cdot \sigma_i \implies T_i^{\text{clean}} = \text{NaN}$$

- The 3σ (99.73% Confidence) Rule:** Under Gaussian error distribution assumptions, 99.73% of valid physical sensor noise falls within $\pm 3\sigma$ of the local rolling mean. Any reading exceeding 3σ represents a noise spike rather than a physical room temperature change.
- Local Rolling Window (24 points):** Using a static global mean would falsely flag the real cooling pulldown drop ($29.9^\circ\text{C} \rightarrow 20.4^\circ\text{C}$) as an outlier. A centered 2-minute rolling window adapts dynamically as the room temperature decreases.

A.3 Linear NaN Interpolation

The NaN values were replaced by linear interpolation between surrounding points (t_k, T_k) and (t_{k+1}, T_{k+1}) :

$$T(t) = T(t_k) + \frac{t - t_k}{t_{k+1} - t_k} (T(t_{k+1}) - T(t_k))$$

A.4 Exponential Moving Average (EMA) Noise Suppression

$$T_{\text{EMA}}(t_k) = \alpha \cdot T(t_k) + (1 - \alpha) \cdot T_{\text{EMA}}(t_{k-1})$$

Where $\alpha = 0.10$ is the smoothing factor.

Appendix B: Model Calibration

B.1 Standard ASHRAE Guideline 14 Calibration Metrics

ASHRAE Guideline 14 (*Measurement of Energy, Demand, and Water Savings*) defines two mandatory statistical metrics for validating building energy models against measured data:

1. Coefficient of Variation of RMSE — CV(RMSE)

$$\text{CV(RMSE)} = \frac{1}{\bar{T}_z} \sqrt{\frac{\sum_{i=1}^N (T_{\text{sim}}(t_i) - T_z(t_i))^2}{N - p}} \times 100\%$$

- **Variables:**
 - $T_{\text{sim}}(t_i)$: EnergyPlus simulated zone temperature at time step i .
 - $T_z(t_i)$: Cleaned weighted average sensor zone temperature at time step i .
 - $\bar{T}_z$: Mean observed sensor zone temperature across test duration ($\bar{T}_z = \frac{1}{N} \sum T_z(t_i)$).
 - N : Number of simulation time steps.
 - p : Number of calibrated model parameters ($p = 4$: k, c_p, ρ, ACH).
- **Physical Meaning & Target:** CV(RMSE) measures relative variance/scatter of simulation errors normalized by mean room temperature.
 - **ASHRAE Standard Target:** $\text{CV(RMSE)} \leq 5.0\%$ for sub-hourly calibrated models.

2. Normalized Mean Bias Error — NMBE

$$\text{NMBE} = \frac{1}{\bar{T}_z} \frac{\sum_{i=1}^N (T_{\text{sim}}(t_i) - T_z(t_i))}{N - p} \times 100\%$$

- **Physical Meaning & Target:** NMBE measures systematic over-prediction or under-prediction bias.
 - Positive NMBE (> 0): EnergyPlus is predicting consistently hotter than reality.
 - Negative NMBE (< 0): EnergyPlus is predicting consistently colder than reality.
 - **ASHRAE Standard Target:** $|\text{NMBE}| \leq 2.0\%$.

B.2 Dynamic Time Warping (DTW) Trajectory Shape Distance

DTW constructs an $N \times N$ pairwise distance matrix $M_{i,j} = |T_{\text{sim}}(t_i) - T_z(t_j)|$ and searches for an optimal warping path $\pi = (\pi_1, \pi_2, \dots, \pi_K)$ that minimizes total path alignment cost:

$$\text{DTW_Distance} = \min_{\pi} \sum_{(i,j) \in \pi} |T_{\text{sim}}(t_i) - T_z(t_j)|$$

Subject to monotonicity and continuity constraints ($i_k - i_{k-1} \leq 1, j_k - j_{k-1} \leq 1$).

- **Why DTW is Essential for AC Pulldown:** DTW accounts for slight phase delays in thermal wall conduction response, penalizing structural shape mismatch (convex vs. concave curves) rather than just static temperature offsets.

B.3 Composite Loss Function & Parameter Normalization

Composite Loss Equation:

Combined ASHRAE Guideline 14 metrics and DTW distance into a single objective loss function:

$$\text{Loss}(\mathbf{x}) = \text{CV}(\text{RMSE}) + 0.5 \cdot |\text{NMBE}| + 2.0 \cdot \text{DTW_Distance}$$

Parameter Normalization to [0, 1]:

We normalized each physical parameter P_j into a unit vector $x_j \in [0, 1]$:

$$P_j(x_j) = P_j^{\min} + x_j \cdot (P_j^{\max} - P_j^{\min})$$

Parameter	$P_j^{\min}$	$P_j^{\max}$	Normalized $x_j = 0.0$	Normalized $x_j = 1.0$
PU Foam Conductivity (k)	0.0150 W/(m · K)	0.0450 W/(m · K)	0.0150	0.0450
PU Foam Specific Heat (c_p)	800.0 J/(kg · K)	1800.0 J/(kg · K)	800.0	1800.0
PU Foam Density (ρ)	20.0 kg/m ³	45.0 kg/m ³	20.0	45.0
Chamber Infiltration (ACH)	0.01 hr ⁻¹	0.50 hr ⁻¹	0.01	0.50

Optimization Technique:

The Bounded Nelder-Mead Simplex algorithm evaluates $\text{Loss}(\mathbf{x})$ on normalized $[0, 1]^4$ space, converting $\mathbf{x}$ back to physical properties $P(\mathbf{x})$ before generating candidate IDFs and running EnergyPlus simulations. Bounded Nelder-Mead was chosen because it is a derivative-free algorithm and requires significantly fewer evaluation runs than metaheuristic methods like Genetic Algorithms.